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EDUCATION

- **Harvard University** Boston, MA
M.S. in Health Data Science August 2025 – March 2027
 - Program requirements complete December 2026 (commencement March 2027); available for full-time start January 2027
 - Coursework: Deep Learning, AI in Medicine, Healthcare Machine Learning, Data Science
- **Emory University, College of Arts and Sciences** Atlanta, GA
B.S. in Applied Mathematics and Statistics; Minor in Computer Informatics; GPA: 3.925 August 2021 – May 2025
 - Coursework: Probability and Statistics, Machine Learning, NLP, Data Structures, Database Systems

TECHNICAL SKILLS

- **Languages:** Python, SQL
- **Backend:** YAML policy compile, Selenium, relational schemas, unittest, pytest
- **Cloud:** AWS EC2, AWS S3, boto3
- **Data:** ETL, AnnData, PDF text extraction
- **Certifications:** AWS Certified Cloud Practitioner, AWS Certified AI Practitioner, AWS Machine Learning Associate

ENGINEERING PROJECTS

- **Human-Supervised Browser Automation System** Boston, MA
Sole author July 2026 – Present
 - Built a job-application system that keeps GitHub as durable policy and evidence, a Google Sheet as live runtime state, and Simplify as the application ledger.
 - Compiles YAML policy into one runtime file the local browser operator reads. Submit is allowed only after listed identity and writing checks.
 - Tests refuse passwords, cookies, OTP, and leaked phone or email. Designed recovery for Mac shutdown, Wi-Fi loss, and browser restart.
- **mixhvg-py: AnnData API and Fidelity Tests** Boston, MA
Sole author July 2026
 - Ported mixhvg to an AnnData-native API (cells $\times$ genes). Each method emits a score, scores become ranks, and each gene keeps its maximum rank.
 - Validated against R / scran / Seurat. Seurat vst and dispersion match at Spearman **1.0000**. The paper-recommended mixture agrees at Jaccard **0.980**.
 - Reproduced two degenerate upstream methods and a Seurat 5 silent-NULl vst bug in tests rather than silently fixing them. 68 pytest tests pass.

RESEARCH EXPERIENCE

- **Computational Legislative Studies: UK and New Zealand Parliamentary Data** Atlanta, GA
Research Assistant — Emory University (Carrubba and Estrada) December 2023 – May 2024
 - Built scrapers and ETL over UK Parliament open data, with a relational schema designed up front. Normalized **22,606** member-level division votes, **4,791** MPs, **4,108** bills, and **3,875** constituencies.
 - Diagnosed Cloudflare 403 challenges and restored collection with TLS-impersonating requests and Selenium. Staged data on AWS EC2 and S3.
 - Wrote a UK-to-New-Zealand transition document for handoff.